

Day 3 — Student Tasks

Part A — Strings

1. Create a variable with a full name (for example `"Ahmed Ali"`). Print how many characters are in that name.
2. Take the sentence `"I am learning JavaScript"`. Print it once in uppercase and once in lowercase.
3. You have an email: `"student@nti.com"`. Check whether this email contains the character `"@"` and print the result.
4. For the word `"JavaScript"`, print the first character and the last character.
5. In the text `"This is a bad day"`, replace the word `"bad"` with `"good"`, then print the new text.
6. Take the sentence `"I love coding"`. Split it into an array of words, then join those words again using `" - "` between them and print the result.
7. You have the text `" nti egypt training "`. Remove the extra spaces from the start and end, convert the text to uppercase, then replace `"EGYPT"` with `"CAIRO"` and print the final result.

Part B — Numbers & Math

8. Convert the string `"45.8"` into a real number, then print that number.
9. Using the number `7.6`, print three results: rounded value, floor value, and ceil value.
10. From the numbers `12, 5, 28, 9`, find and print the largest number and the smallest number.
11. Generate a random integer between 1 and 20 (inclusive) and print it.
12. Take the price `19.4567` and print it with only 2 digits after the decimal point.
13. Generate 5 random integers between 1 and 50. Then print the largest number, the smallest number, and the average of the 5 numbers (show the average with 2 decimal places).

Part C — Loops

14. Write a `for` loop that prints all numbers from 1 to 20.

15. Write a `for` loop that prints only the odd numbers from 1 to 15.
16. Write a `while` loop that counts down from 10 to 1 and prints each number.
17. You have this array: `["Sara", "Omar", "Mona", "Youssef"]`. Use `for...of` to print each name on a new line.
18. Use a `for` loop with `break` to print numbers from 1 to 10, but stop the loop before printing 7. The output should be: 1 2 3 4 5 6
19. Use nested loops to print the multiplication table from 1 to 3. Each line should look like: `2 * 3 = 6`
20. Write a program that prints all numbers from 1 to 30, but: skip every number that is divisible by 3 using `continue`, and stop completely when you reach number 25 using `break`.

Part D — Mix (Strings + Numbers + Loops)

21. Using a loop, print each letter of the word `"HELLO"` on a separate line.
22. You have the array `[10, 20, 30, 40]`. Calculate the sum of all numbers and print the total.
23. Count how many times the letter `"a"` or `"A"` appears in this sentence: `"JavaScript is amazing and awesome"`. Then print the count.
24. From the grades array `[70, 85, 92, 60, 77, 88]`, print only the even grades.
25. Using nested loops, print this stars pattern:

```
*
* *
* * *
* * * *
```
26. You have this array of student names: `["ahmed", "sara", "omar", "laila", "hassan"]`. Loop through the array, convert each name to uppercase, then print only the names that start with the letter `"A"` or `"S"`, and also print how many names matched.